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Thermochemical waste-heat recuperation as on-board hydrogen production technology

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HIGHLIGHTS

- The efficiency of TCR systems is affected by operating parameters.
- The maximum efficiency is 600 K for methanol; 700 K for ethanol and 900 K for methane.
- The maximum recuperation rate is 0.58 for TCR system by steam methane reforming.
- The maximum transformation coefficient observed for steam methane reforming.

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ABSTRACT

Thermochemical waste-heat recuperation (TCR) as an on-board hydrogen production technology is considered. To determine the effectiveness of using TCR systems as an on-board hydrogen production technology and to assess the possibility of hydrogen production in TCR systems, a thermodynamic analysis of various hydrocarbon reforming reactions was carried out. The thermodynamic analysis has been realized via Aspen HYSYS software. Three steam reforming reactions with methane, methanol, and ethanol were investigated. It was established that the composition of the initial reaction mixture and the process temperature has a significant effect on the efficiency of the thermochemical heat recuperation system. The maximum efficiency of thermochemical heat recuperation systems due to steam reforming is achieved at 600 K for methanol; 700 K for ethanol and 900 K for methane.

Introduction

The use of hydrogen as a fuel is an important scientific and technical task. In recent years, many works on the use of hydrogen in transport, aviation, industrial furnaces, etc. have been published. The main advantages of hydrogen as a fuel are as follows: clean energy without pollution; most abundant element in the Universe; lightest fuel with richest in energy per unit mass; direct conversion into thermal, mechanical, and electrical energy [1–3]. At the same time, due to the lack of infrastructure and the difficulties of transporting and storing hydrogen, hydrogen is used very little as a fuel [4].

One of the intermediate stages of using hydrogen as a fuel is the production of hydrogen or hydrogen-rich fuels directly at the facility in which it will be used as a fuel. This hydrogen technology is called on-board hydrogen production [5].

The traditional approach in large-scale hydrogen production is based on the use of various reforming hydrocarbons processes with steam, carbon dioxide, etc [6–9]. A distinctive feature of this method of hydrogen and synthesis gas (hydrogen-rich fuels) production is the need to supply heat to

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